# Triggers and Basic SQL Queries in DBMS

**What is a Trigger?**

- A trigger is a special stored procedure in Database Management Systems.
- It automatically executes when certain events occur in a table.
- Triggers work automatically without user action.

**Definition**

**Trigger:**
A database object that automatically executes in response to INSERT, UPDATE, or DELETE operations.

**Why Triggers are Used?**

1. Maintain data integrity
2. Automatically update tables
3. Keep audit records
4. Improve security
5. Enforce business rules

**Events That Activate Triggers**

| **Event** | **Meaning** |
|---|---|
| INSERT | When new row is added |
| UPDATE | When data is modified |
| DELETE | When row is removed |

**Types of Triggers**

1. BEFORE Trigger
2. AFTER Trigger
3. INSTEAD OF Trigger

**1. BEFORE Trigger**

**Definition**

- Executes before INSERT, UPDATE, or DELETE operation.

**Example**

```
CREATE TRIGGER Check_Age
BEFORE INSERT
ON Student
FOR EACH ROW
BEGIN
  IF NEW.Age < 18 THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT='Age must be 18 or above';
  END IF;
END;
```

**Explanation**

**Step-by-Step**

1. Trigger name → Check_Age
2. Executes BEFORE INSERT
3. Checks Age value
4. If Age < 18:
    - Error message shown
5. Invalid data rejected

**Important Points**

6. Used for validation
7. Prevents wrong data insertion

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**2. AFTER Trigger**

**Definition**

- Executes after operation is completed.

**Example**

```
CREATE TRIGGER Student_Log
AFTER INSERT
ON Student
FOR EACH ROW
BEGIN
  INSERT INTO LogTable
  VALUES(NEW.Student_ID,'Record Inserted');
END;
```

**Explanation**

**Step-by-Step**

1. New student inserted
2. Trigger activates automatically
3. Log entry stored in LogTable

**Important Points**

8. Used for auditing
9. Keeps history of changes

### 3. DELETE Trigger

**Example**

```
CREATE TRIGGER Delete_Log
AFTER DELETE
ON Student
FOR EACH ROW
BEGIN
  INSERT INTO LogTable
  VALUES(OLD.Student_ID,'Record Deleted');
END;
```

**Explanation**

10. OLD keyword stores deleted values
11. Trigger records deletion activity

**NEW and OLD Keywords**

| Keyword | Meaning |
|---|---|
| NEW | New inserted/updated value |
| OLD | Existing/deleted value |

**General Syntax of Trigger**

```
CREATE TRIGGER trigger_name
BEFORE/AFTER INSERT/UPDATE/DELETE
ON table_name
FOR EACH ROW
BEGIN
  SQL statements;
END;
```

**Components of Trigger**

12. Trigger Name
13. Event
14. Table Name
15. Action Body

**Advantages of Triggers**

16. Automatic execution
17. Improves integrity
18. Maintains logs
19. Enforces business rules
20. Reduces manual work
21. Improves security
22. Maintains consistency

**Disadvantages of Triggers**

23. Difficult debugging
24. May reduce performance
25. Hidden execution logic

**Real-Time Example**

**Banking System**

**Trigger Use**

- Automatically update account balance after transaction.

BASIC SQL QUERIES

**What is SQL Query?**

- SQL query is a command used to interact with database tables.

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**Types of Basic SQL Queries**

1. SELECT
2. INSERT
3. UPDATE
4. DELETE

**1. SELECT Query**

**Definition**

- Retrieves data from table.

**Syntax**

```
SELECT column_name
FROM table_name;
```

**Example**

```
SELECT * FROM Student;
```

**Output**

| Student_ID | Name | Age |
|---|---|---|
| 101 | Ravi | 20 |

**Important Points**

26.* means all columns

27.Most used query

**2. INSERT Query**

**Definition**

- Adds new records.

**Syntax**

INSERT INTO table_name
VALUES(values);

**Example**

INSERT INTO Student
VALUES(102,'Anu',21);

**Important Points**

28.Adds new rows

29.Values should match columns

**3. UPDATE Query**

**Definition**

- Modifies existing data.

**Syntax**

UPDATE table_name
SET column=value
WHERE condition;

**Example**

```
UPDATE Student
SET Age=22
WHERE Student_ID=102;
```

**Important Points**

30. Changes existing values
31. WHERE clause is important

**4. DELETE Query**

**Definition**

- Removes records.

**Syntax**

```
DELETE FROM table_name
WHERE condition;
```

**Example**

```
DELETE FROM Student
WHERE Student_ID=102;
```

**Important Points**

32. Deletes selected rows
33. WHERE prevents deleting all records

**Additional Important Points**

34. Triggers execute automatically.

35. SQL queries are user commands.
36. INSERT adds data.
37. UPDATE modifies data.
38. DELETE removes data.
39. SELECT retrieves data.
40. BEFORE trigger executes before operation.
41. AFTER trigger executes after operation.
42. NEW keyword stores new values.
43. OLD keyword stores old values.
44. Triggers improve automation.
45. Queries manipulate database data.
46. Triggers maintain consistency.
47. SQL is used in relational databases.
48. Triggers support auditing.
49. Basic queries are important in DBMS.
50. Triggers reduce manual checking.

**Short Definitions for Exams**

**Trigger**

A database object that automatically executes when specific events occur.

**SQL Query**

A command used to retrieve or manipulate database data.

**One-Mark Important Points**

- Triggers execute automatically.
- BEFORE trigger executes before event.
- AFTER trigger executes after event.

- SELECT retrieves data.
- INSERT adds records.